

## Robotics-for-Cacao Barter

Follow-up action item · Agroverse · 2026-08-06

**Context.** Four problems facing cacao farming: witches' broom disease, an aging farmer population with no labor to replace it, inconsistent roasting, and farm-succession risk that threatens agroforestry continuity. A robotics design expert partner is on board. Working-prototype reference: **Frasky**, the IIT vineyard robot (Reuters video, <https://youtu.be/hg8qYrjyYCU>). This follow-up turns that introduction into a concrete pilot.

### 1 · The four problems

#### Problem 1 — Witches' broom (*vassoura-de-bruxa*, *Moniliophthora perniciosa*)

The single highest-leverage canopy disease in Bahia cacao: infected brooms must be **cut off and collected — physically removed from the plantation** in dry season, or spores reinfect — cutting yields by 30–50% on affected trees. Detection and cutting are manual, repetitive, and easy to miss. **No chemical spray:** this is an agroforestry system, so witches' broom is managed by mechanical removal, not treatment.

#### Problem 2 — Aging farmer population + labor shortage

Cacao farmers are aging, and there is **no younger generation stepping in** — farm management and harvest depend on manual labor that is increasingly hard to find and afford. The physical work (climbing, cutting, pod-breaking, carrying) is heavy, seasonal, and unattractive to new workers. **The strain and injury risk are what eventually force an aging owner to stop** — this is the direct on-ramp to Problem 4. The farm's know-how is concentrated in aging owners who cannot scale it.

#### Problem 3 — Roasting consistency

Roasting — the step that defines the flavor profile — is a **sequential decision process with delayed reward**: the operator acts on the roast curve (heat, airflow, drum speed, time), and only *after* the batch do you learn whether the flavor profile landed. Today that is skilled-craft, tuned batch-by-batch by experienced tasters — **not scalable** and hard to reproduce consistently across batches, roasters, and operators.

#### Problem 4 — Farm succession risk (the existential one)

When a farmer is hurt, or simply **too old**, the farm gets sold. There is **no guarantee the new owner continues agroforestry** — they may clear the land, plant something else, or run it down. Every farm lost to succession is hectares of regenerated rainforest **permanently reversed**, directly threatening the **10,000-hectare regeneration goal**. The agroforestry system lives or dies with the individual owner — that is fragile.

### 2 · Working prototype reference — Frasky (IIT)

**Video (Reuters):** <https://youtu.be/hg8qYrjyYCU> — "A hands-on robot tends grapes in Italy's vineyards"

- Built by the **Italian Institute of Technology + Bergamo-region agricultural stakeholders** (roboticist Dr. Manuel Catalano).

- Autonomously **monitors grape clusters** (camera + digital map of every cluster's location), **manipulates plants** (robotic arm), and **applies targeted spray treatments**.
- Designed for precision agriculture, **labor-shortage relief**, and sustainability — field-tested in vineyards (Reuters, Nov 2025).

**Proof point:** a permanent tree/vine crop's canopy work can be robotized — detection, manipulation, precision. Cacao is the same category.

**One deliberate difference — no spray.** Frasky applies targeted chemical spray. Agroverse's system is **agroforestry**: we do **not** use chemical treatments. The cacao adaptation is **cut & collect** — the robot cuts off infected brooms and collects them off the plantation, so spores cannot reinfect. The Frasky value we borrow is the *mechanical* capability (monitor → manipulate → act), not the chemical one.

| Frasky capability (grape)       | Cacao adaptation                                                            | Problems it addresses          |
|---------------------------------|-----------------------------------------------------------------------------|--------------------------------|
| Cluster monitoring via camera   | Broom / pod detection via vision                                            | P1 broom, P2 labor             |
| Manipulate plants (robotic arm) | <b>Cut &amp; collect</b> infected brooms (mechanical removal, no chemicals) | P1 broom, P2 labor             |
| Targeted spray treatments       | <b>Not adopted</b> — agroforestry, no chemical inputs                       | —                              |
| —                               | Selective pod <i>harvest</i> (varied heights: trunk + branches)             | P2 labor                       |
| —                               | Farm-management data (what to do, where, when)                              | P2 aging owners, P4 succession |

### 3 · Roasting consistency as a reinforcement learning problem

The reframe — this is a reinforcement learning problem.

| RL element                 | In the cacao roaster                                                                                      |
|----------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>State (observation)</b> | Bean moisture, bean temperature / rate-of-rise, roast color, particle size distribution, volatile markers |
| <b>Action</b>              | Heat input, airflow, drum speed, batch timing — the roast-curve decisions                                 |
| <b>Reward</b>              | Flavor-profile match (sensor + taste panel), batch-to-batch consistency, energy efficiency                |
| <b>Policy</b>              | A learned roast curve that reproduces a target flavor profile, batch after batch                          |

The agent **learns the roast policy** over batches — each batch is a training episode, the reward comes at the end. That is exactly what makes it RL rather than a static recipe: the controller adapts to bean lot, moisture, and machine state to hit the flavor target.

**The observation layer — particle / roast-state detection.** The RL agent is only as good as what it can sense:

| Detection approach           | What it measures                                                          | Status                      |
|------------------------------|---------------------------------------------------------------------------|-----------------------------|
| Machine vision (camera)      | Bean color, surface browning, particle size distribution                  | ■ proven in food lines      |
| Hyperspectral / SWIR imaging | Moisture, roast degree, bean defects (slaty, moldy, under/over-fermented) | ■ proven, conveyor-scalable |
| Portable NIR spectroscopy    | Chemical markers of roast degree, in-line                                 | ■ proven, low-cost          |
| Electronic nose + ANN        | Volatile profile → predicted roast degree (94.4% accuracy reported)       | ■ proven                    |

**Why now / why us.** Precedent exists (IMA Group's "AI Learning to Roast", model-based roast optimization) but is sold as expensive industrial software. A robotics + ML partner can build a **low-cost RL roast controller + particle detector** tuned to the beans and flavor profiles the Agroverse network actually runs — turning roast flavor control from craft into a **scalable, self-improving process**.

#### 4 · The proposal (barter model)

Offer robotics + automation services to Bahia/Pará farmers **in exchange for cacao as payment**:

| Role                        | Who                                   |
|-----------------------------|---------------------------------------|
| Robot design & build        | Robotics design expert (Gianluca)     |
| Farmer network & deployment | Agroverse — Bahia operations (Ilhéus) |
| Market for cacao received   | Agroverse (US import live)            |

**The four problems → four workstreams:**

| Problem                    | Robotic/AI response                                                                                        | Feasibility                         |
|----------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------|
| P1 · Witches' broom        | Broom detection + <b>cut &amp; collect</b> robot (mechanical removal, no chemicals — agroforestry-aligned) | ■ tractable now                     |
| P2 · Aging farmers / labor | Harvest assist + farm-management data (what/where/when)                                                    | ■ tractable (harvest ■■ harder R&D) |
| P3 · Roast consistency     | RL roast controller + particle detection                                                                   | ■ parallel, lower-field-risk track  |
| P4 · Succession risk       | <b>Automation extends ownership longevity</b> (below)                                                      | ■ structural, low-tech-risk         |

**Why P4 matters most.** P1–P3 improve the farm. P4 **keeps the farm from being lost**. Robotics + the barter agreement together can make an agroforestry cacao farm *more valuable as agroforestry* than as cleared land — the strongest possible protection for regenerated hectares.

### The core causal chain — automation extends farmer ownership longevity:

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Automation takes over the heavy, high-injury work

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Less physical strain on the aging farmer — fewer injuries, less exhaustion

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The farmer can keep running the farm safely and productively, years longer

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Fewer forced sales from injury or "too old to work" — ownership longevity extended

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Farms stay in agroforestry hands → succession risk drops

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Regenerated hectares stay standing → 10,000-hectare goal holds

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The root cause of the forced sale is often **injury or labor that is simply too hard** for an older body.

Automation directly attacks that root cause: the robot does the climbing, cutting, and carrying — the farmer supervises and manages. It is not just "less work"; it is **more years of safe, productive ownership** — which is exactly what keeps agroforestry land from changing hands and being cleared.

**Honest MVP scoping.** Broom *detection + cut & collect* (mechanical removal, no chemicals) is tractable now — it is the robot's arm doing what farm workers do today, but tireless and precise. *Selective cacao pod harvesting* is harder R&D — set expectations so the pilot stays deliverable.

## 5 · Pilot plan (1 season, measured)

1. **Scoping call with the robotics partner** — share the **Frasky video** (<https://youtu.be/hg8qYrjyYCU>) as the reference prototype; define MVP build, season timing, and cost basis; evaluate the roast-RL track (state sensors, reward signal, data capture per batch).
2. **Farmer pitch (PT-BR)** — value prop: labor shortage, aging-owner relief, **fewer injuries and less strain so the owner can keep farming longer**, broom yield loss, **no cash outlay**, paid in cacao at fair market BRL; **no chemicals** (agroforestry-compatible); succession protection (the farm stays productive and valuable as cacao agroforestry).
3. **2–3 test farms** in the Bahia (Ilhéus) network.
4. **Barter accounting** — value cacao received at market BRL/USD and record it into the supply ledger at fair market value; draft the agroforestry-continuity clause for the agreement.
5. **Measure, then scale** — % broom removed, harvest kg, labor-hours saved, roast batch-to-batch consistency, **farmer-reported strain/injury reduction, years of continued ownership (delayed succession), farms retained in agroforestry across succession events**. No scaling on unmeasured evidence.

## 6 · Next steps (owners)

- **Approve outreach copy** and introduce the robotics partner (WhatsApp).
- **Draft scoping-call question list** (incl. Frasky feasibility + roast-RL questions: observation sensors, reward definition, data per batch) and the PT-BR farmer pitch.
- **Robotics partner** — feasibility call; MVP definition.
- **Agroverse Brazil operations** — identify 2–3 test farms; coordinate season logistics.

## 7 · Tie-back to mission

Every broom cut and collected, every robot-assisted harvest, every aging farmer given a tool instead of a shovel — and every batch of cacao roasted to a reproducible flavor profile — keeps more trees producing and more **Amazon rainforest standing. Automation that keeps the farmer safely owning and farming longer is automation that keeps agroforestry land out of the clearing market.** When a farm finally does pass to new hands, the robotics + cacao-market system makes **continuing agroforestry the profitable choice** — that is how we reach and hold **10,000 hectares regenerated.**

*Draft v11 · pending approval before any outreach.*